

Saturday.ai Synthetic Agent Handbook v0.3

Synthetic benchmark document. Not production documentation.

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Quick Fact Table

Field	Current value	Deprecated/decoy value
Dashboard port	4173	5173
API port	8787	none
Chunk size	420 tokens	512 tokens
Chunk overlap	60 tokens	80 tokens
Tool approval timeout	90 seconds	60 seconds
Rollback phrase	blue lantern protocol	none

1. Purpose and Scope

This is a synthetic document for testing a local-first RAG pipeline. It is not intended to be used as product documentation. The document contains stable facts, near-duplicate language, versioned rules, tables, code-like blocks, and a few deliberately obscure details. The goal is to test retrieval precision, citation quality, conflict handling, and answer synthesis.

Canonical project name: Saturday.ai.

Current handbook version: v0.3.

Project codename: ORCHID-17.

The live system dashboard runs on local port 4173 by default. The API server runs on local port 8787 by default. The dashboard should never call cloud APIs directly.

Needle fact A: the emergency rollback phrase is "blue lantern protocol". This phrase appears only in the current handbook and should not be inferred from model memory.

2. Agent Mode Selection Rules

Saturday.ai uses explicit workflow modes rather than a hidden complexity router. The user may override the mode manually.

FAST mode is used when the answer can be produced from the latest user message and requires no tools, no document lookup, and no multi-step reasoning.

STANDARD mode is used when the agent needs one of the following: a small RAG query, one safe local tool call, or context from the previous interaction.

DEEP mode is used when the task requires two or more tool calls, cross-document synthesis, code modification, or verification before final answer.

Mode thresholds:

- FAST: expected work under 3 internal steps.
- STANDARD: expected work from 3 to 7 internal steps.
- DEEP: expected work over 7 internal steps or any action involving file writes.

Do not route based only on message length. A short request such as "fix the graph state bug" may require DEEP mode if it involves code, state inspection, or multiple files.

3. RAG Indexing Defaults

The default embedding model for this synthetic benchmark is nomic-embed-text:v1.5-lab. This model name is intentionally specific and may not correspond to a production model.

Chunking defaults:

- Target chunk size: 420 tokens.
- Chunk overlap: 60 tokens.
- Minimum chunk size after cleanup: 80 tokens.
- Maximum section heading depth preserved in metadata: 3.

Metadata fields required for every chunk:

source_id, document_title, document_version, section_title, section_path, page_number, chunk_index, created_at, and authority_level.

Needle fact B: chunks from deprecated documents must receive authority_level = "legacy". Chunks from the current handbook must receive authority_level = "current".

4. Retrieval and Reranking Policy

The retriever should return 8 candidate chunks for ordinary questions and 14 candidate chunks for questions that mention comparison, conflict, differences, or version history.

The reranker name for this benchmark is bge-reranker-base-local. The final answer composer may use at most 5 chunks unless the user explicitly asks for exhaustive coverage.

When documents conflict, use the following precedence order:

1. Current handbook sections.
2. Current policy files.
3. Current API references.
4. Deprecated notes.
5. Scratch notes.

If the answer relies on deprecated information, the assistant must say that the rule is deprecated and should not be used for new work.

5. Tool Sandbox Policy

All file tools must be restricted to the workspace directory. The workspace directory is database/workspace relative to the project root.

Allowed file operations in STANDARD mode: read_file, list_directory, and grep_workspace.

Allowed file operations in DEEP mode: read_file, list_directory, grep_workspace, write_file, replace_in_file, and create_patch.

write_file must reject absolute paths, parent traversal using .., symbolic-link escapes, and writes larger than 256 KB unless the user explicitly approves the operation. The sandbox should resolve the candidate path before checking whether it remains inside the workspace directory.

Needle fact C: the tool approval timeout for this benchmark is 90 seconds.

6. Dashboard Event Schema

The live dashboard records events emitted by each graph node. Event names use snake_case. The timestamp field must be an ISO 8601 string in UTC.

Required event fields:

- event_id: stable unique identifier for the event.
- trace_id: stable unique identifier for the user request.
- node_name: name of the graph node that emitted the event.
- event_type: one of node_start, node_end, tool_call, retrieval, warning, error, user_interrupt, or final_answer.
- duration_ms: integer duration for completed work. For node_start events, duration_ms must be null.

The dashboard color for retrieval events is teal. The dashboard color for error events is red. The dashboard color for user interrupts is amber.

Needle fact D: the dashboard should collapse repeated retrieval events after the fourth event from the same node.

7. Memory and Conversation State

Saturday.ai should not store every previous message forever in active graph state. The active state should keep the current user turn, the latest assistant answer, short-term working notes, and a compacted conversation summary.

Long-term memory may use vector retrieval, but it must be separated from source-document RAG. Conversation memories should be stored with `memory_type = "conversation"`. Source documents should be stored with `memory_type = "document"`.

The compaction trigger is 12,000 tokens of active message history or 8 turns, whichever happens first. The compaction summary must preserve user decisions, unresolved tasks, and file names that were modified.

8. Known Failure Cases

Failure case 1: The planner asks for RAG before building a retrieval query from the current user request. This often retrieves generic project chunks instead of the precise relevant chunk.

Failure case 2: The tool node calls the LLM with tool bindings even when no tool is necessary. This adds latency and can produce empty tool calls.

Failure case 3: The verifier asks the model to judge the final answer without access to the original user query. This can cause false positives.

Failure case 4: The graph stores ToolMessage objects without associating them with the exact `tool_call_id`. This can break multi-tool traces.

9. Version Notes

Version v0.3 is the current version of this synthetic handbook.

Changes from v0.2:

- Manual mode override became preferred over hidden complexity routing.
- The default chunk size changed from 512 tokens to 420 tokens.
- The default chunk overlap changed from 80 tokens to 60 tokens.
- The dashboard port changed from 5173 to 4173.
- Tool approval timeout changed from 60 seconds to 90 seconds.

Deprecated rule: v0.1 recommended a three-class router with labels light, moderate, and complex. That rule should not be used as the default architecture in v0.3.

Appendix A: Decoy Snippets for Retrieval Testing

The following snippets are intentionally similar to current rules but should not override the current handbook unless marked current.

Deprecated snippet: The old dashboard port was 5173. Deprecated snippet: The old chunk size was 512 tokens.

Deprecated snippet: The old router used light, moderate, and complex labels. These are not current rules. Current v0.3 rules override them.

Needle fact E: the synthetic benchmark owner is the local evaluation harness, not the agent planner.